Core Pure 2 · Chapter 3

Methods in Calculus

Improper integrals · Mean values · Inverse trigonometric functions · Partial fractions

Chapter Overview 2

Improper integrals

Replace an infinite limit or an undefined endpoint with a limit.

Check convergence before evaluating.

Further techniques

Find the mean value of a function.

Differentiate and integrate inverse trigonometric functions.

Use partial fractions in integration.

1. Why does direct substitution give an impossible result for $\int_{-1}^{1} x^{-2}\,dx$?

1

2. Evaluate $\int_{1}^{\infty} x^{-2}\,dx$.

2

Undefined endpoints

Evaluate $\int_{0}^{1} x^{-1/2}\,dx$.

Endpoint limit

Infinite intervals

An integral can converge even though its interval is unbounded. For example, the normal density has total area 1:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz = 1$$

Always check that each required limit exists.

Examples 5

Compare $\int_1^\infty x^{-2}\,dx$ and $\int_1^\infty x^{-1}\,dx$.

Compare

A function tending to zero is not sufficient for its integral to converge.

Similarly, the harmonic series $\sum_{r=1}^\infty 1/r$ diverges despite its terms tending to zero.

When f(x) not defined for some value 6

1. Evaluate or establish divergence: $\int_0^1 x^{-2}\,dx$.

1

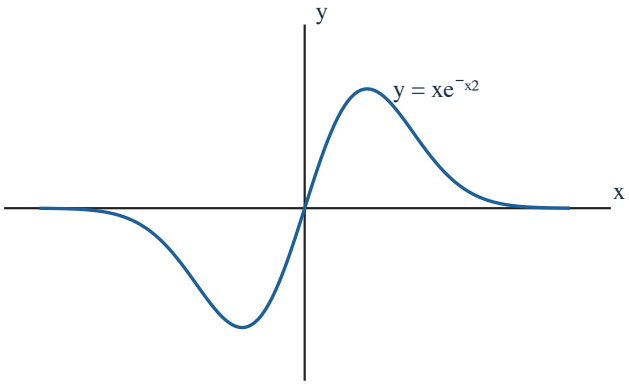
2. Evaluate $\int_0^2 x/\sqrt{4-x^2}\,dx$.

2

An improper integral with two infinite limits 7

Evaluate $\int_{-\infty}^{\infty} xe^{-x^2}\,dx$.

Split at zero; both integrals must converge separately.



Integral

Exercise 3A - Core Pure Year 2 Pages 56-58 8

5a. Find $\int (7-3x)^{-2}\,dx$.

b. Hence evaluate the integral from $-\infty$ to 2.

a

b

6. Find $\int x^2 e^{x^3}\, dx$, then evaluate $\int_{-\infty}^{1} x^2 e^{x^3}\, dx$.

Integrate

Limit

7. Find $\int \ln x / x\, dx$. Does $\int_{1}^{\infty} \ln x / x\, dx$ converge?

Integrate

Convergence

8. Integrate $(\ln x)^2$. Hence evaluate the integral over $(0, 1]$ and discuss $(0, \infty)$.

Integrate

Limits

9. Evaluate $\int_0^2 6x(4 - x^2)^{-1/3}\, dx$.

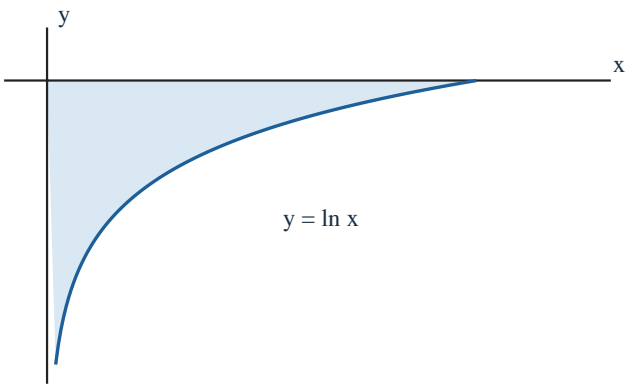
9

10. Evaluate

$$\int_{-2}^{2} \frac{\sqrt{2 - x} - 3\sqrt{2 + x}}{\sqrt{4 - x^2}}\, dx$$

10

11. Find the finite area bounded by $y = \ln x$, the x-axis and the y-axis. You may use $x \ln x \to 0$ as $x \to 0^+$.



11

12a. Explain why $\int_0^{\pi/2} \tan x \, dx$ is improper.

b. Determine whether it converges.

a

b

The Mean Value of a Function 15

Mean value of a function

The discrete mean is $(y_1 + \cdots + y_n)/n$. For equally spaced samples, a limiting average gives:

Continuous mean

Examples 16

1. Find the mean of $4/\sqrt{2+3x}$ on $[2, 6]$.

1

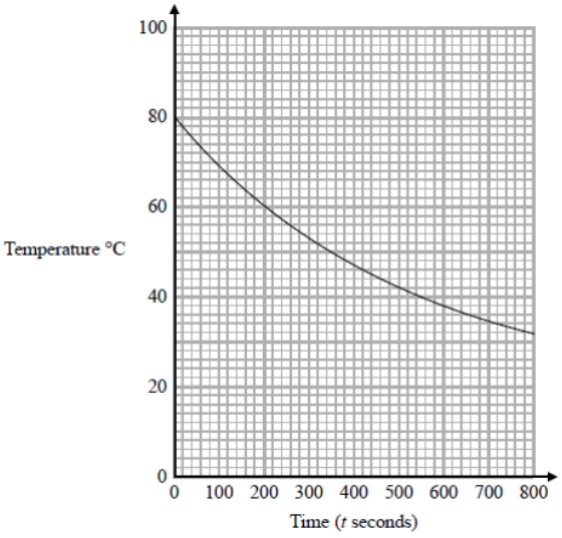
2. For $f(x) = 4/(1+e^x)$, find the means of f, $f+4$, and $-f$ on $[\ln 2, \ln 6]$.

2: integrate

2: means

Why does the gradient joining the first and last points give the average rate of change?

Apply this to water cooling from $80°\text{C}$ between $t = 0$ and $t = 800$.



Average gradient

4. Find the exact mean of

$$f(x) = \frac{\sin x \cos x}{\cos 2x + 2}, \qquad 0 \leq x \leq \pi/2$$

4

6. Find the exact mean of $f(x) = x \sin 2x$ on $[0, \pi/3]$.

6

7. Let $f(x) = 5x/((2x-1)(x+2))$.

a. Show its mean on $[1, 5]$ is $\frac{1}{4}\ln(49/3)$.

b. Find the mean of $f(x) + \ln k$, $k > 0$, in the form $p \ln q$.

a

b

8. Let $f(x) = x(x^2 - 4)^4$.

a. Show its mean on $[0, 2]$ is $256/5$.

b. Hence find the mean of $-2f(x)$.

a

b

11. Find the exact mean of $f(x) = 1/\sqrt{2 - x}$ over $[0, 2]$.

11

13. Let $f(x) = \cos x/(2+\sin x)^2$.

a. Integrate f.

b. Find its mean over $[0, 5\pi/3]$.

c. Find the mean of $f(x) + 3x$ over that interval.

a

b

c

Differentiating inverse trigonometric functions 24

Show that $d(\arcsin x)/dx = 1/\sqrt{1-x^2}$.

Derivation

$$\frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$$

Differentiate $y = \arcsin(x^2)$.

Chain rule

Test Your Understanding 25

1. Differentiate
$y = \arctan((1-x)/(1+x))$.

1

2. Differentiate $y = \operatorname{arcsec}(2x)$, for $x > 1/2$.

2

Exercise 3C - Core Pure Year 2 Pages 64-65 26

3. Differentiate
$$y = \frac{1 + \arctan x}{1 - \arctan x}$$

3

4. Prove that $\arccos x + \arcsin x = \pi/2$ for $-1 \leq x \leq 1$.

4

6. Given $\tan y = x \arctan x$, find dy/dx in terms of x.

6

7. For $y = \arcsin x$, prove $(1 - x^2)y'' - xy' = 0$.

7

Exercise 3C - Core Pure Year 2 Pages 64-65 29

a. $y = \arcsin(\arcsin x)$ b. $y = \arccos(\arccos x)$ c. $y = \arctan(\arctan x)$

a

b

c

a graph

b graph

c graph

Integrating with inverse trigonometric functions 30

Use a substitution to prove

$$\int \frac{1}{1+x^2} dx = \arctan x + C$$

Substitution

Integrate

Integrals leading to inverse sine

31

For $a > 0$ and $|x| < a$, show

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin(x/a) + C$$

Proof

Example

32

Standard results

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin(x/a) + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan(x/a) + C$$

Here $a > 0$; the first formula requires $|x| < a$.

Find $\int 4/(5 + x^2)\, dx$.

Example

Inverse trigonometric integrals: practice 33

1. Find $\int 1/(25+9x^2)\,dx$.

1

2. Evaluate

$$\int_{-\sqrt{3}/4}^{\sqrt{3}/4} \frac{dx}{\sqrt{3-4x^2}}$$

2

Final example 34

Find

$$\int \frac{x+4}{\sqrt{1-4x^2}}dx$$

First term

Second term

Combined

4. Find constants A, B such that

$$\int \frac{dx}{4+3x^2} = A\arctan(Bx) + C$$

4

5. Find constants P, Q such that

$$\int \frac{dx}{\sqrt{3-4x^2}} = P\arcsin(Qx) + C$$

5

7. Evaluate $\int_{\sqrt{2}}^{\sqrt{3}} (4 - x^2)^{-1/2}\, dx$.

7

8. Find $\int (2 + 3x)/(1 + 3x^2)\, dx$.

8

9. Integrate $(2x - 1)/\sqrt{2 - x^2}$ in the form $A \arcsin(x/\sqrt{2}) + B\sqrt{2 - x^2} + C$.

9

Exercise 3D - Core Pure Year 2 Pages 66-67 39

11. Find $\int (4x-1)/\sqrt{6-5x^2}\,dx$.

First term

Second term

Combined

Challenge 3D - Core Pure Year 2 Pages 66-67 40

Challenge. For $x > 1$, use $x = \sec\theta$ to integrate:

a. $1/(x\sqrt{x^2-1})$

b. $\sqrt{x^2-1}/x$.

Substitute

a

b

Solving using partial fractions 41

Use partial fractions to prove, for $a > 0$,

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a + x}{a - x} \right| + C$$

Split

Integrate

Partial Fractions involving Quadratic Factors 42

Quadratic factors

Use a general linear numerator over an irreducible quadratic:

$$\frac{1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$$

Find $\int (1 + x)/(x^3 + 9x)\, dx$.

Partial fractions

Integral

Test Your Understanding 43

a. Express $(x^4 + x)/(x^4 + 5x^2 + 6)$ in partial fractions.

b. Hence integrate it.

Equal numerator and denominator degrees require a constant quotient.

a

b

Exercise 3E - Core Pure Year 2 Pages 72-73 44

2. Find $\int (x^2 - 3x)/((x^2 + 6)(x + 2))\,dx$ in the form $A \ln|x + 2| + B \arctan(x/\sqrt{6}) + C$.

Split

Integrate

3. Let $f(x) = x^4 - x^3 - 4x^2 - 2x - 12$.

a. Given $x + 2$ is a factor, fully factorise f.

b. Integrate $(x^3 - 20x^2 + 4x - 24)/f(x)$.

a

b

4. Find $\int (2 - x)/(x^3 + 4x)\, dx$ in the form $A \ln(x^2/(x^2 + 4)) + B \arctan(x/2) + C$.

Split

Integrate

6. Integrate $(x^3 + 9x^2 + x + 1)/(x^4 - 1)$, giving a logarithm of a quotient plus $D \arctan x$.

Split

Integrate

7. Let $f(x) = x^3 - 4x^2 + 6x - 24$.

a. Given $f(4) = 0$, factorise f.

b. Express $(2x^2 - 3x + 24)/f(x)$ in partial fractions.

c. Hence integrate it.

a

b

c

8. Let $f(x) = 1/((x-2)(2x-1))$.

a. Integrate f.

b. Show $\int_{1/2}^{2} f(x)\,dx$ diverges.

a

b

9. Express $(x^4 + 5x^2 + 2x)/(x^4 + 10x^2 + 24)$ in partial fractions, then integrate it.

a

b

10. Use partial fractions to integrate $(x^2 + 4x + 10)/(x^3 + 5x)$, for $x > 0$.

Split

Integrate

11. Evaluate $\int_0^2 2/((x+1)(x^2+1))\,dx$.

Check carefully whether the claimed value $(\pi + 2\ln 2)/4$ matches this upper limit.

Integrate

Evaluate

12. Express $(x^4+1)/(x(x^2+2)^2)$ in partial fractions, then integrate it.

a

b